

JOBSHEET STUDI KASUS 2



Disusun oleh:

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1. Kode:

Class Pembeli:

```
package JobsheetCM2.Model;

public class Pembeli10 {
    public String namaPembeli;
    public String noHp;
    public int noAntri;

    public Pembeli10(int noAntri, String nama, String no) {
        namaPembeli = nama;
        noHp = no;
        this.noAntri = noAntri;
    }

    public void tampilPembeli() {
        System.out.printf("%-15d %-10s %-12s\n", noAntri, namaPembeli, noHp);
    }
}
```

Class Pesanan:

```
package JobsheetCM2.Model;

public class Pesanan10 {
    public int kodePesanan;
    public String namaPesanan;
    public int harga;

    public Pesanan10(int kode, String namaPesanan, int harga) {
        this.kodePesanan = kode;
        this.namaPesanan = namaPesanan;
        this.harga = harga;
    }
}
```

Class Node10:

```
package JobsheetCM2.Model;

public class Node10 {
    public Pembeli10 dataPembeli;
    public Pesanan10 dataPesanan;
    public Node10 prev;
    public Node10 next;
```

```

public Node10() {}

public Node10(Pembeli10 dataPembeli) {
    this.dataPembeli = dataPembeli;
    this.prev = null;
    this.next = null;
}

public Node10(Pesanan10 dataPesanan) {
    this.dataPesanan = dataPesanan;
}
}

```

Class AntrianPembeli:

```

package JobsheetCM2.Model;

public class AntrianPembeli10 {
    public Node10 head;
    public Node10 tail;

    public AntrianPembeli10() {
        head = null;
        tail = null;
    }

    public boolean isEmpty() {
        return head == null;
    }

    public void addLast(Pembeli10 data) {
        Node10 newNode = new Node10(data);
        if (isEmpty()) {
            head = tail = newNode;
        } else {
            tail.next = newNode;
            newNode.prev = tail;
            tail = newNode;
        }
    }

    public void print() {
        if (isEmpty()) {
            System.out.println("Antrian masih kosong\n");
            return;
        }
    }
}

```

```

Node10 current = head;

while (current != null) {
    Pembeli10 currentPembeli = current.dataPembeli;
    System.out.printf("%-15d %-10s %-12s\n", currentPembeli.noAntri,
currentPembeli.namaPembeli, currentPembeli.noHp);
    current = current.next;
}
}

public void removeFirst() {
    if (isEmpty()) {
        System.out.println("Antrian masih kosong.\n");
        return;
    }

    if (head == tail) {
        head = tail = null;
        System.out.println("Data Antrian Pembeli berhasil dihapus.");
    } else {
        head = head.next;
        head.prev = null;
        System.out.println("Data Antrian Pembeli berhasil dihapus.");
    }
}
}

```

Class AntrianPesanan:

```

package JobsheetCM2.Model;

public class AntrianPesanan10 {
    public Node10 head;
    public Node10 tail;

    public AntrianPesanan10() {
        head = null;
        tail = null;
    }

    public boolean isEmpty() {
        return head == null;
    }

    public void addLast(Pesanan10 data) {

```

```

Node10 newNode = new Node10(data);
if (isEmpty()) {
    head = tail = newNode;
} else {
    tail.next = newNode;
    newNode.prev = tail;
    tail = newNode;
}
}

public void print() {
    if (isEmpty()) {
        System.out.println( "Antrian masih kosong\n");
        return;
    }

    Node10 current = head;

    sortByNamaPesanan();

    while (current != null) {
        Pesanan10 currentPesanan = current.dataPesanan;
        System.out.printf("%-15d %-15s %-10d\n", currentPesanan.kodePesanan,
currentPesanan.namaPesanan, currentPesanan.harga);
        current = current.next;
    }
}

public void removeFirst() {
    if (isEmpty()) {
        System.out.println("Antrian masih kosong.\n");
        return;
    }

    if (head == tail) {
        head = tail = null;
        System.out.println("Data Antrian berhasil dihapus.\n");
    } else {
        head = head.next;
        head.prev = null;
        System.out.println("Data Antrian berhasil dihapus.\n");
    }
}

public void sortByNamaPesanan() {
    Node10 current = head;

```

```

Node10 currNext = null;
Pesanan10 temp;

while (current != null) {
    currNext = current.next;

    while (currNext != null) {
        int resultCompare =
current.dataPesanan.namaPesanan.compareTo(currNext.dataPesanan.namaPesanan);

        if (resultCompare > 0) {
            temp = current.dataPesanan;
            current.dataPesanan = currNext.dataPesanan;
            currNext.dataPesanan = temp;
        }

        currNext = currNext.next;
    }
    current = current.next;
}
}
}

```

Class View:

```

package JobsheetCM2.View;

import java.util.Scanner;

import JobsheetCM2.Model.*;

public class View10 {
    public int cetakMenu(Scanner sc, int pil) {
        System.out.println("=====");
        System.out.println("SISTEM ANTRIAN ROYAL DELISH");
        System.out.println("=====");
        System.out.println("1. Tambah Antrian");
        System.out.println("2. Cetak Antrian");
        System.out.println("3. Hapus Antrian dan Pesan");
        System.out.println("4. Laporan Pesanan");
        System.out.println("0. Keluar");
        System.err.print("Pilih menu: ");
        pil = sc.nextInt();
        sc.nextLine();
        return pil;
    }
}

```

```

int noAntri = 1;
public void tambahAntriPembeli(Scanner sc, AntrianPembeli10 antrian) {
    System.out.print("Nama Pembeli : ");
    String nmPembeli = sc.nextLine();
    System.out.print("No HP      : ");
    String no = sc.nextLine();
    Pembeli10 pembeli = new Pembeli10(noAntri, nmPembeli, no);
    antrian.addLast(pembeli);
    noAntri++;

    System.out.println("Antrian berhasil ditambahkan dengan nomor: " +
antrian.tail.dataPembeli.noAntri + "\n");
}

public void tambahAntriPesanan(Scanner sc, AntrianPesanan10 antrian) {
    System.out.println("=== TAMBAH PESANAN ===");
    System.out.print("Kode Pesanan : ");
    int kd = sc.nextInt();
    sc.nextLine();
    System.out.print("Nama Pesanan : ");
    String nm = sc.nextLine();
    System.out.print("Harga      : ");
    int harga = sc.nextInt();
    sc.nextLine();
    Pesanan10 pesanan = new Pesanan10(kd, nm, harga);
    antrian.addLast(pesanan);
}

public void cetakAntrian(AntrianPembeli10 antrian) {
    System.out.println("=====");
    System.out.println("Daftar Antrian Pembeli");
    System.out.println("=====");

    if (!antrian.isEmpty()) {
        System.out.printf("%-15s %-10s %-12s\n", "No Antrian", "Nama", "No HP");
        antrian.print();
        System.out.println();
    } else {
        System.out.println("Antrian masih kosong.\n");
    }
}

public void hapusNPesan(Scanner sc, AntrianPembeli10 antrianPembeli,
AntrianPesanan10 antrianPesanan) {
    if (antrianPembeli.isEmpty()) {

```

```

        System.out.println("Antrian masih kosong.\n");
    }

    String nmPembeli = antrianPembeli.head.dataPembeli.namaPembeli;

    antrianPembeli.removeFirst();

    tambahAntriPesanan(sc, antrianPesanan);
    String nmPesanan = antrianPesanan.tail.dataPesanan.namaPesanan;
    System.out.println(nmPembeli + " telah memesan " + nmPesanan + ".\n");
}

public void laporanPesanan(AntrianPesanan10 antrian) {
    System.out.println("=====");
    System.out.println("LAPORAN PESANAN (URUT NAMA PESANAN)");
    System.out.println("=====");

    if (!antrian.isEmpty() && antrian.head.dataPesanan != null) {
        System.out.printf("%-15s %-15s %-10s\n", "Kode Pesanan", "Nama Pesanan",
"Harga");
        antrian.print();
        System.out.println();
    } else {
        System.out.println("Laporan masih kosong.\n");
    }
}
}
}

```

Class SistemAntrianKasir:

```
package JobsheetCM2;
```

```
import java.util.Scanner;
```

```
import JobsheetCM2.View.*;
```

```
import JobsheetCM2.Model.*;
```

```
public class SistemAntrianKasir10 {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        AntrianPembeli10 antrianBeli= new AntrianPembeli10();
```

```
        AntrianPesanan10 antrianPesan = new AntrianPesanan10();
```

```
        View10 view = new View10();
```

```
        int pil = 0;
```



```
do {
    pil = view.cetakMenu(sc, pil);

    switch (pil) {
        case 1:
            view.tambahAntriPembeli(sc, antrianBeli);
            break;

        case 2:
            view.cetakAntrian(antrianBeli);
            break;

        case 3:
            view.hapusNPesan(sc, antrianBeli, antrianPesan);
            break;

        case 4:
            view.laporanPesanan(antrianPesan);
            break;

        case 0:
            System.out.println("Keluar.");
            break;

        default:
            System.out.println("Pilihan tidak valid.\n");
            break;
    }
} while (pil != 0);

sc.close();
}
```

2. Output:

```
=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu: 1
Nama Pembeli   : Vania
No HP          : 084222234556
Antrian berhasil ditambahkan dengan nomor: 4

=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu: 2
=====
Daftar Antrian Pembeli
=====
No Antrian      Nama      No HP
1               Ainra     082245000000
2               Danra     082245111111
3               Sanri     082245222222
4               Vania     084222234556
```

```

=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu: 3
Data Antrian Pembeli berhasil dihapus.
=== TAMBAH PESANAN ===
Kode Pesanan : 123
Nama Pesanan : Es Teler
Harga : 5000
Danra telah memesan Es Teler.

```

```

=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu: 3
Data Antrian Pembeli berhasil dihapus.
=== TAMBAH PESANAN ===
Kode Pesanan : 124
Nama Pesanan : Mie Goreng
Harga : 34000
Sanri telah memesan Mie Goreng.

```

```

=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu: 2
=====
Daftar Antrian Pembeli
=====
No Antrian      Nama      No HP
4               Vania     084222234556

```

```

=====
SISTEM ANTRIAN ROYAL DELISH
=====
1. Tambah Antrian
2. Cetak Antrian
3. Hapus Antrian dan Pesan
4. Laporan Pesanan
0. Keluar
Pilih menu: 4
=====
LAPORAN PESANAN (URUT NAMA PESANAN)
=====
Kode Pesananan  Nama Pesanan  Harga
125             Es Degan      6000
123             Es Teler      5000
124             Mie Goreng    34000

```